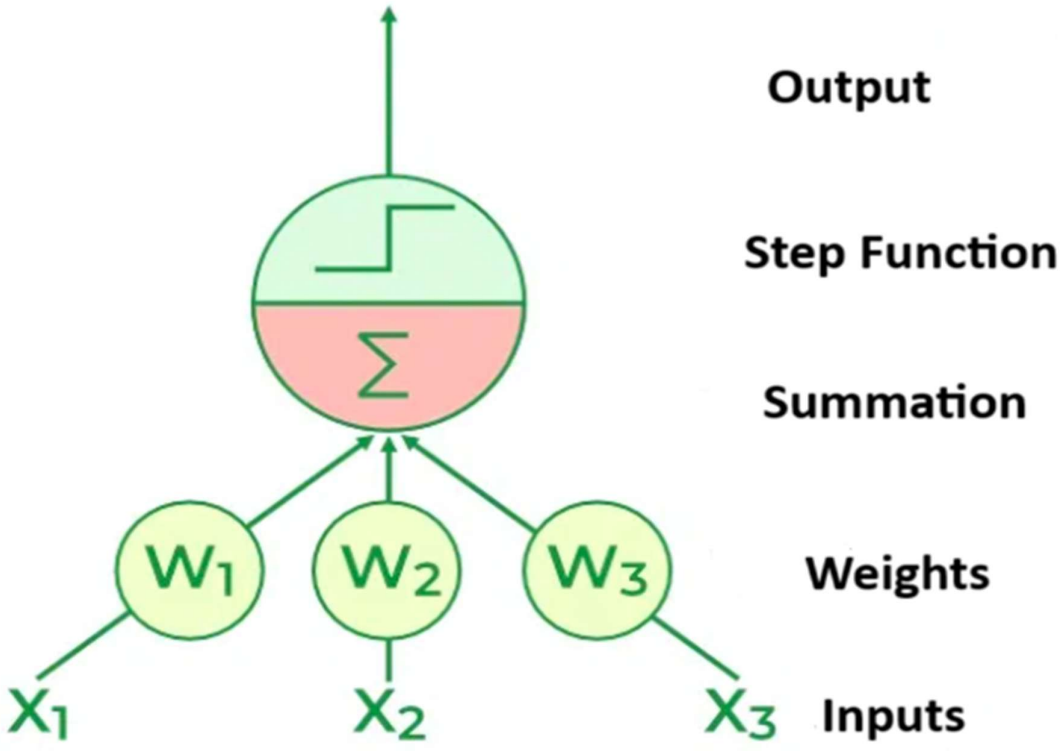
To Study and Implement Single Layer Perceptron for Binary Classification

The aim of this experiment is to implement AND and OR logic gates using a Single Layer Perceptron to showcase the basic capabilities of a neural network in binary decision-making.

Introduction

The perceptron is one of the simplest artificial neural network architectures, introduced by Frank Rosenblatt in 1957. It represents the simplest type of feedforward neural network, consisting of a single layer of input nodes that are fully connected to a layer of output nodes. The perceptron can learn linearly separable patterns and uses slightly different types of artificial neurons known as threshold logic units (TLU). The TLU was first introduced by McCulloch and Walter Pitts in the 1940s. A perceptron has a single layer of threshold logic units with each TLU connected to all inputs.



Perceptron Learning Rule:

- The perceptron learning rule updates the weights based on the error signal.
- If the predicted output matches the target output, no weight adjustment is made.
- If the predicted output differs from the target output, the weights are adjusted to reduce the error.
- The learning rate, which determines the step size of weight updates, is an important parameter that affects the convergence and stability of the model.
- Equation for perceptron weight adjustment :

$$\Delta w = \eta * d * x$$

where,
d : Predicted Output – Desired Output,
η : Learning Rate, Usually Less than 1,
x : Input Data.

Applications:

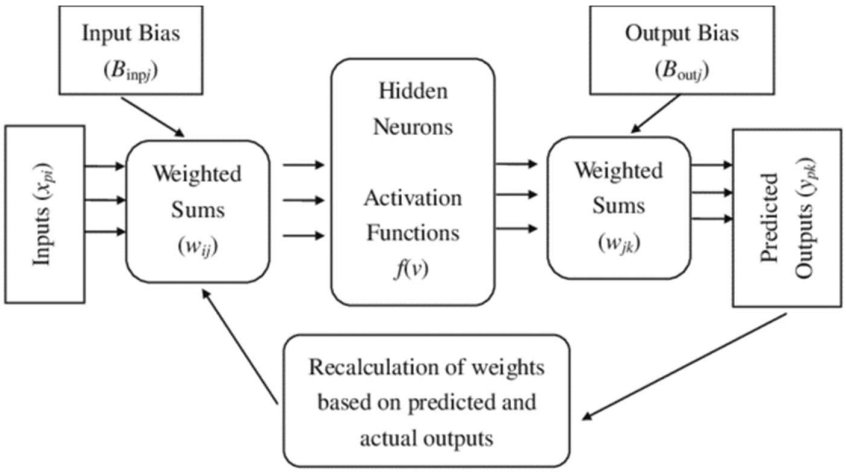
- **Binary Classification :** SLPs find application in tasks requiring binary classification, such as spam detection, medical diagnosis, and quality control.
- **Logical Operations :** SLPs can be used to implement simple logical operations such as AND, OR, and NOT. For example, a perceptron can be trained to perform an AND operation by classifying input pairs as either belonging to the class "1" (true) or "0" (false) based on the logical AND condition.
- **Fault Detection in Systems :** SLPs can be applied to detect faults or anomalies in systems, where the goal is to classify normal and abnormal conditions based on input data.

Limitations:

- **Linear Separability :** One significant limitation is that SLPs can only learn linearly separable functions, meaning they struggle with problems that require non-linear decision boundaries.
- **Exclusive OR (XOR) Problem :** SLPs fail to solve problems like the XOR function, which demands a non-linear separation of input classes.

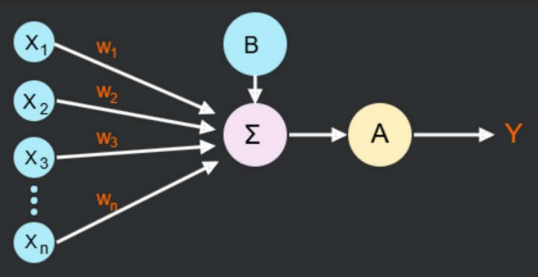
<u>Training Algorithm:</u>

- The training algorithm for the single-layer perceptron is based on the concept of supervised learning.
- The perceptron algorithm adjusts the weights iteratively to minimize the error between the predicted output and the actual output.
- The algorithm updates the weights using the delta rule, which involves calculating the difference between the predicted and actual outputs and adjusting the weights accordingly.



SELECT GATE | COMPUTE | ANALYZE | INSTRUCTIONS

NETWORK



CONTROLS

Select the GATE : AND SUBMIT

TERMINOLOGY

X_i: The values corresponding to Variables i.e. X_1 , X_2 , ... X_n

W_i: The values corresponding to Weights i.e. W_1 , W_2 , ... W_n

B: The values corresponding to Bias

Σ: The Summation Value ($\Sigma = x_1w_1 + x_2w_2 + x_3w_3 + ... + x_nw_n$)

A: The Activation Function (step function)

Y: The Output

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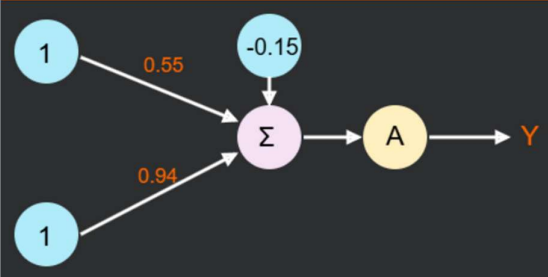
SELECT GATE | COMPUTE | ANALYZE | INSTRUCTIONS

AND NETWORK



CONTROLS

b : -0.15

X_1 : W_1 : 0.55

X_2 : W_2 : 0.94

RESET SUBMIT CHECK UPDATE NEXT

INITIALIZE ACTIVATE

OBSERVATION TABLE FOR EPOCH 1

S. No.	X_1	X_2	T	Y_{in}	Y	ΔW_1	ΔW_2	Δb	W_1	W_2	b
1	-1	-1	-1	-1.40	-1	0	0	0	0.79	0.7	0.09
2	-1	1	-1	-0.00	-1	0	0	0	0.79	0.7	0.09
3	1	-1	-1	0.18	1	-0.24	0.24	-0.24	0.55	0.94	-0.15
4	1	1	1	1.34	1	0	0	0	0.55	0.94	-0.15

CALCULATION

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